Lynbrook Math Summer Camp 2026
G5 Coordinates and 3D

Lynbrook Math Club

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G5.1 Lecture

While previous lectures focused on synthetic geometry (angles, similar triangles, and area formulas), many competition problems are best solved by placing shapes on a coordinate plane or extending our 2D logic into three dimensions. In this section, we will explore analytical geometry (coordinates) and strategies for tackling 3D geometry problems.

Coordinates in 2D

Placing a geometry problem on the Cartesian plane allows us to use algebra to find lengths, areas, and intersections. The foundation of this relies on equations of lines. The most common forms are slope-intercept ($y = mx + b$) and standard form ($Ax + By = C$). Finding the intersection of two lines simply requires setting their equations equal or solving the resulting system of equations.

An important property to remember is that if two non-vertical lines are perpendicular, their slopes multiply to -1. That is, if line L_1 has slope m_1 and L_2 has slope m_2, they are perpendicular if and only if $m_1 \cdot m_2 = -1$.

Distance from a Point to a Line

Often, we need to find the shortest distance (the perpendicular distance) from a point (x_0, y_0) to a line given in standard form $Ax + By + C = 0$. The formula is:

$$d = \frac{|Ax_0 + By_0 + C|}{\sqrt{A^2 + B^2}}$$

Example 1

Find the shortest distance from the point $(3, 4)$ to the line $3x - 4y + 12 = 0$.

Solution: Plugging our point and line coefficients into the formula gives:

$$d = \frac{|3(3) - 4(4) + 12|}{\sqrt{3^2 + (-4)^2}} = \frac{|9 - 16 + 12|}{\sqrt{9 + 16}} = \frac{5}{5} = \mathbf{1}$$

Pick's Theorem

For polygons whose vertices all lie on exact grid points (integer coordinates), Pick's Theorem provides a beautiful shortcut for finding area. Let I be the number of interior grid points and B be the number of grid points exactly on the boundary of the polygon. The area A is:

$$A = I + \frac{B}{2} - 1$$

Example 2

A triangle on the coordinate plane has vertices at $(0, 0)$, $(4, 0)$, and $(2, 3)$. What is its area?

Solution: We can count the boundary and interior points. The boundary points are $(0, 0), (1, 0), (2, 0), (3, 0), (4, 0)$ on the base, plus $(2, 3)$ at the top, and no other integer points lie exactly on the slanted sides. So, $B = 6$. The interior points are $(2, 1), (2, 2), (1, 1)$, and $(3, 1)$. So, $I = 4$. Using Pick's Theorem:

$$A = 4 + \frac{6}{2} - 1 = 4 + 3 - 1 = \mathbf{6}$$

(Checking with the standard $A = \frac{1}{2}bh$, we get $\frac{1}{2}(4)(3) = 6$, which matches perfectly!)

Shoelace Theorem

If a polygon is not easily solved by Pick's Theorem (for instance, its vertices are fractions, or the grid points are too massive to count), the Shoelace Theorem allows you to calculate the area of any polygon given the coordinates of its vertices.

List the coordinates vertically in counterclockwise order, repeating the first coordinate at the end. Then, multiply diagonally downwards and add, multiply diagonally upwards and add, take the absolute difference of these two sums, and divide by 2.

Example 3

Find the area of a pentagon with vertices at $(0, 0)$, $(4, 0)$, $(5, 3)$, $(2, 5)$, and $(-1, 2)$.

Solution: We list the vertices, repeating the first at the end:

$$\begin{array}{cc} 0 & 0 \\ 4 & 0 \\ 5 & 3 \\ 2 & 5 \\ -1 & 2 \\ 0 & 0 \end{array}$$

Sum of downwards diagonals ($x_i \cdot y_{i+1}$): $(0)(0) + (4)(3) + (5)(5) + (2)(2) + (-1)(0) = 0 + 12 + 25 + 4 + 0 = 41$.

Sum of upwards diagonals ($y_i \cdot x_{i+1}$): $(0)(4) + (0)(5) + (3)(2) + (5)(-1) + (2)(0) = 0 + 0 + 6 - 5 + 0 = 1$.

The area is half the absolute difference:

$$A = \frac{1}{2}|41 - 1| = \mathbf{20}$$

A Note on Strategy: In general, you should try to use coordinates as a *last resort*. Synthetic geometry (using angles, lengths, and pure geometric theorems) is often faster and less prone to messy algebra. Furthermore, just because you are using coordinates doesn't mean you can't use other techniques! Similar triangles, the Pythagorean theorem, and circle properties often pair perfectly with coordinate geometry to save you time.

3D Geometry Questions

Inevitably, these types of questions do occur a fair amount of times on the AMC and other competitions. The idea in general here is to try to use symmetry, or other techniques such as taking the proper cross-section, to reduce the problem to a lower-dimensional or very symmetric one. It is very rare that you will actually need to visualize complex 3D shapes purely in your head; instead, look for ways to project them into 2D planes.

Using Cross-Sections

Taking a slice of a 3D figure creates a flat 2D shape that is much easier to work with.

Example 4

A sphere of radius 13 is intersected by a flat plane. The shortest distance from the center of the sphere to the plane is 5. What is the area of the circular cross-section created by this intersection?

Solution: Instead of thinking in full 3D, take a 2D cross-section that cuts perfectly through the center of the sphere and is perpendicular to the intersecting plane. This reduces the sphere to a simple circle of radius 13. The intersecting plane becomes a line (a chord of this circle). The distance from the center of the circle to the chord is 5.

Drawing a radius to the endpoint of the chord creates a right triangle with hypotenuse 13 and leg 5. By the Pythagorean theorem (or recognizing the $5-12-13$ triple), the other leg is 12. This leg is the radius of our cross-sectional circle.

The area of the cross-section is $\pi r^2 = \pi(12^2) = \mathbf{144}\pi$.

Using Symmetry and Unfolding

Sometimes, a 3D problem can be flattened into a 2D problem by "unfolding" or "unrolling" the 3D shape into its flat geometric net.

Example 5

An ant is positioned at one vertex of a solid wooden cube of side length 6. It wants to crawl along the surface of the cube to reach the exact opposite vertex. What is the length of the shortest possible path?

Solution: It is tempting to think the shortest path goes directly along the edges $(6+6+6=18)$. To find the true shortest path, we can use symmetry and reduce the dimensions by "unfolding" the two faces the ant travels across into a flat 2D plane.

If we unfold two adjacent square faces so they lay flat side-by-side, they form a 6×12 rectangle. The starting vertex and ending vertex are now opposite corners of this 6×12 rectangle. The shortest distance between two points on a flat surface is a straight line.

By the Pythagorean theorem:

$$d = \sqrt{6^2 + 12^2} = \sqrt{36+144} = \sqrt{180} = \mathbf{6\sqrt{5}}$$

Since $6\sqrt{5} \approx 13.4$, this straight-line path across the flat net is indeed much shorter than walking along the edges!